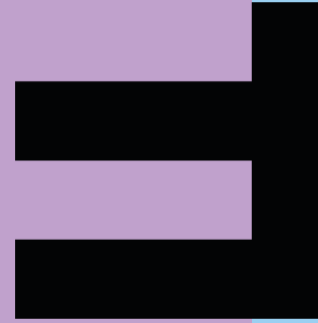


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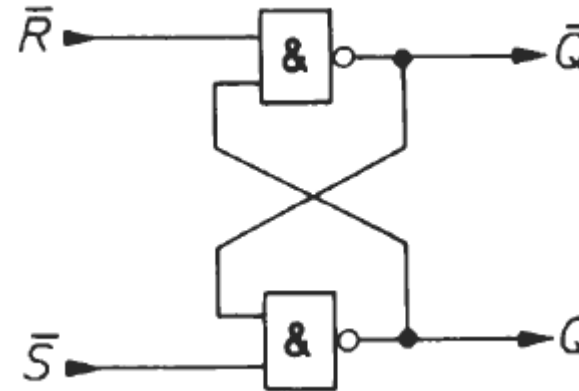
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A Simple Latch (Flip-flop)

- Why latches (flip-flops)?
- The RS latch (flip-flop):



[1], p. 686

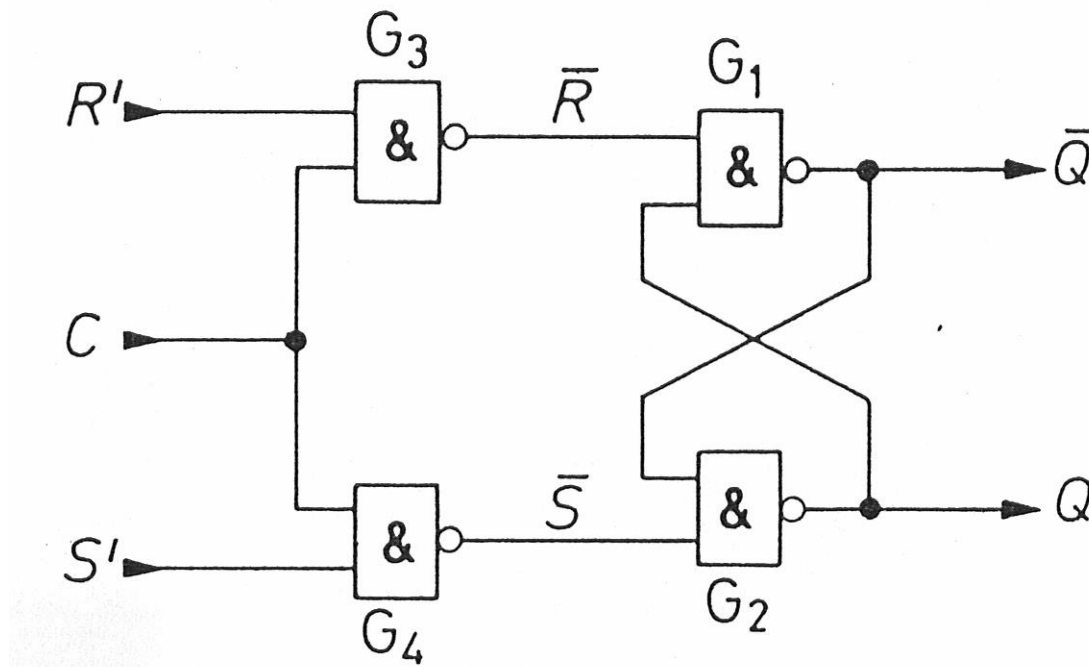
$\bar{S}$	$\bar{R}$	Q	$\bar{Q}$
0	0	(1)	(1)
0	1	1	0
1	0	0	1
1	1	Q_{-1}	$\bar{Q}_{-1}$



RS Latch (Flip-flop) with Clock

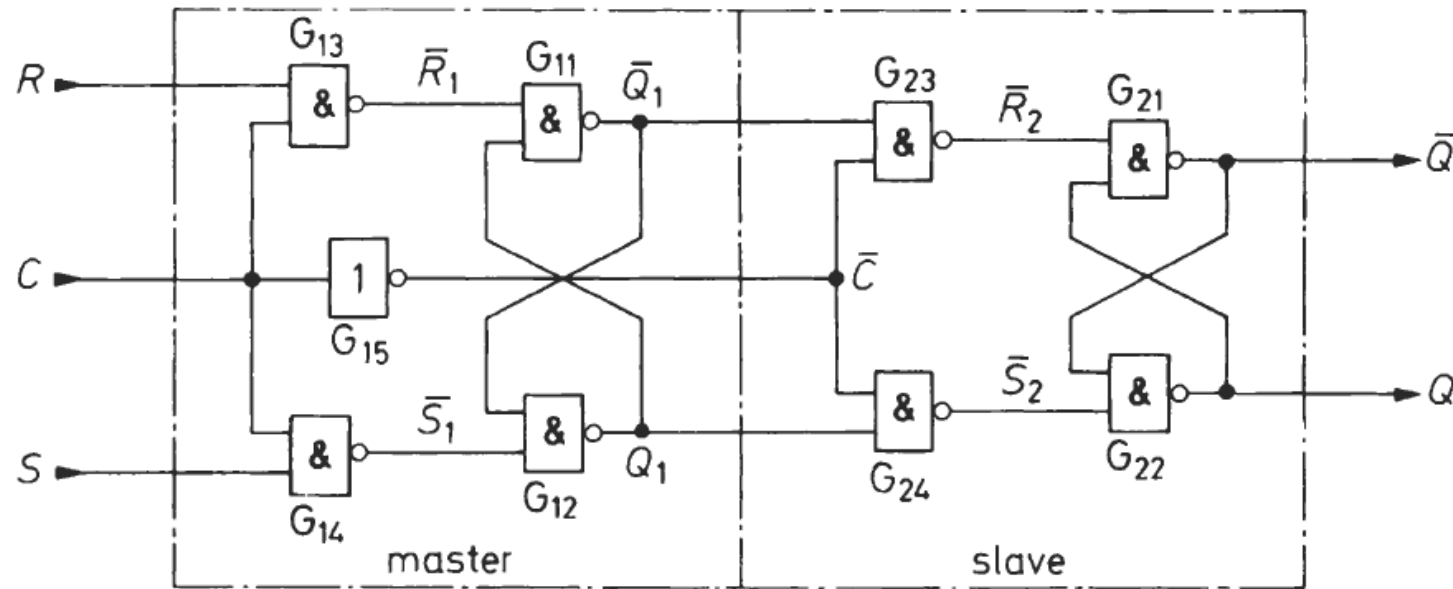
- Why a clock?
- The RS latch (flip-flop) with clock:

[1], p. 686



The Master-slave Flip-flop

- Why master-slave?
- The RS master-slave flip-flop:



[1], p. 688

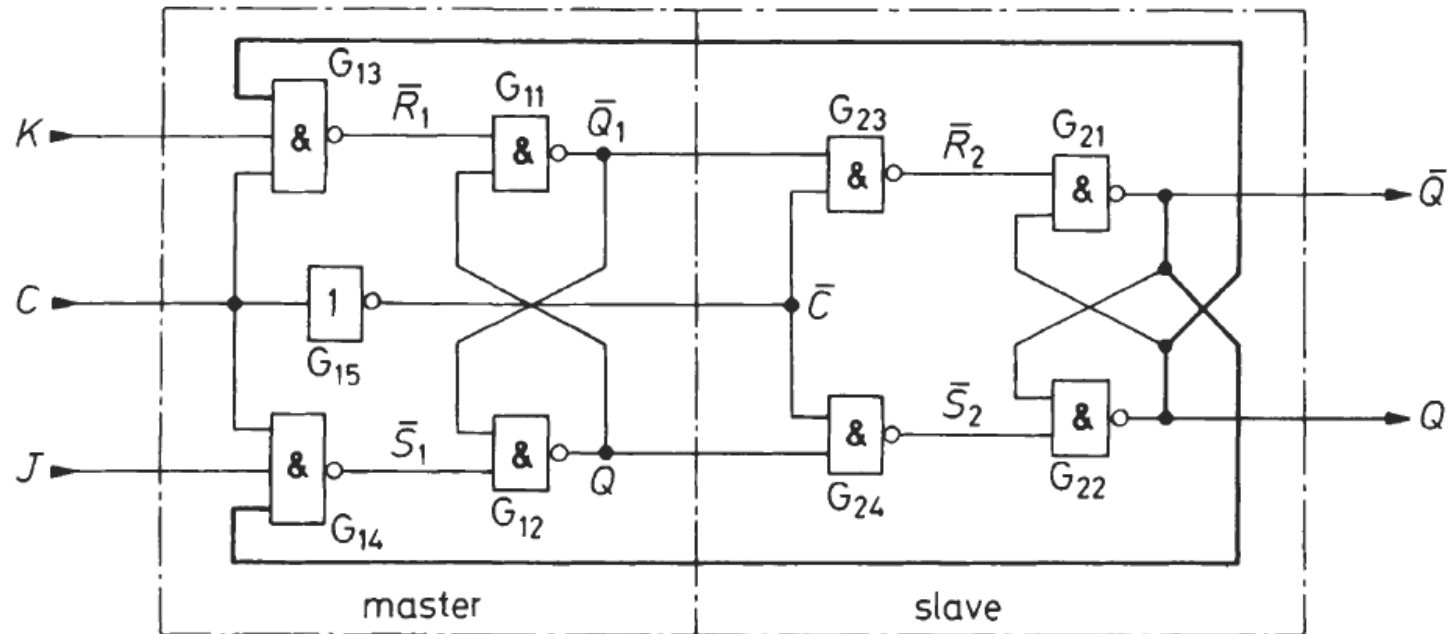


The JK Flip-flop

- Why a JK Flip-flop?
- The JK Flip-flop:

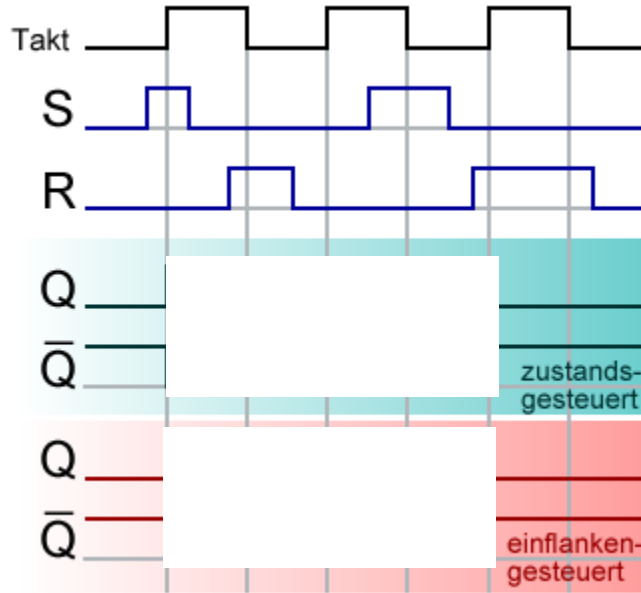
[1], p. 689

J	K	Q
0	0	Q_{-1} (unverändert)
0	1	0
1	0	1
1	1	$\bar{Q}_{-1}$ (invertiert)



Recap: Latch vs. Flip-flop

- Level triggered versus edge triggered:

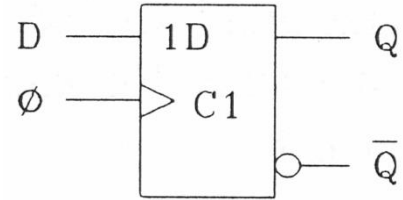


- During the active phase (clock level) of a level triggered latch, any changes of the input can be observed at the output => transparent latch.



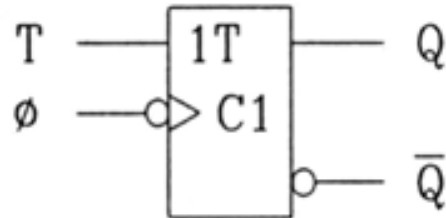
Utilization of the JK-FF

- The D Flip-flop:



Application: Register

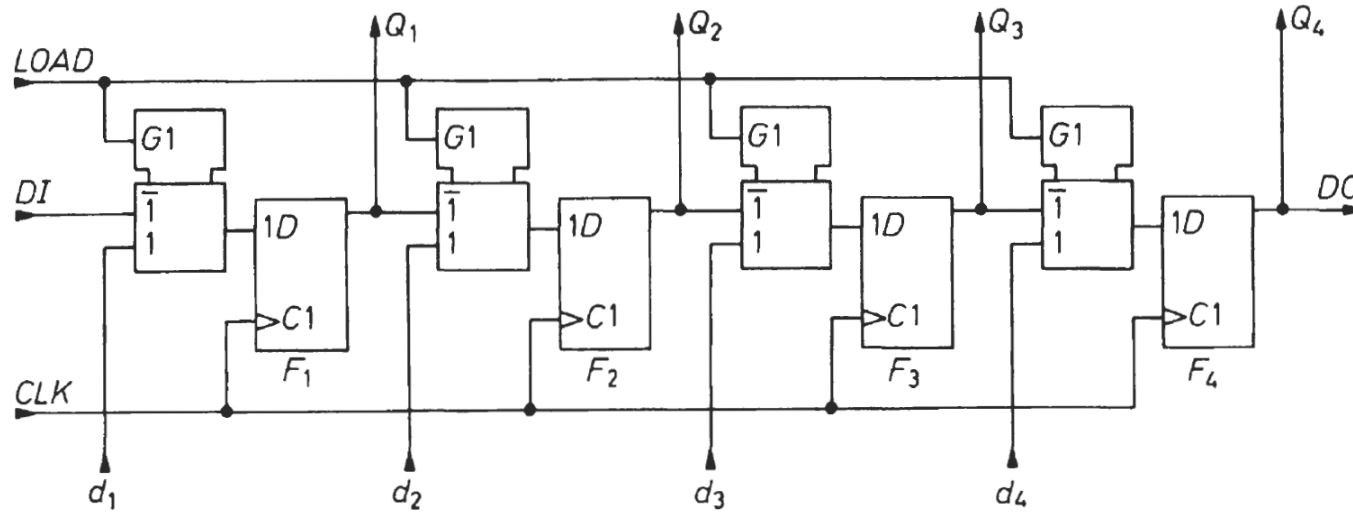
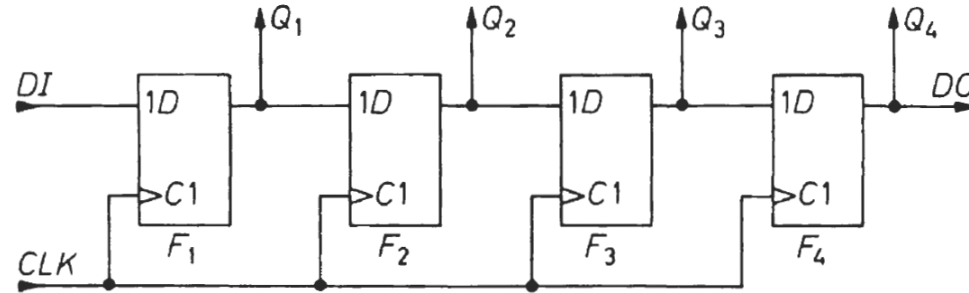
- The T Flip-flop:



Application: Counter



The Shift Register



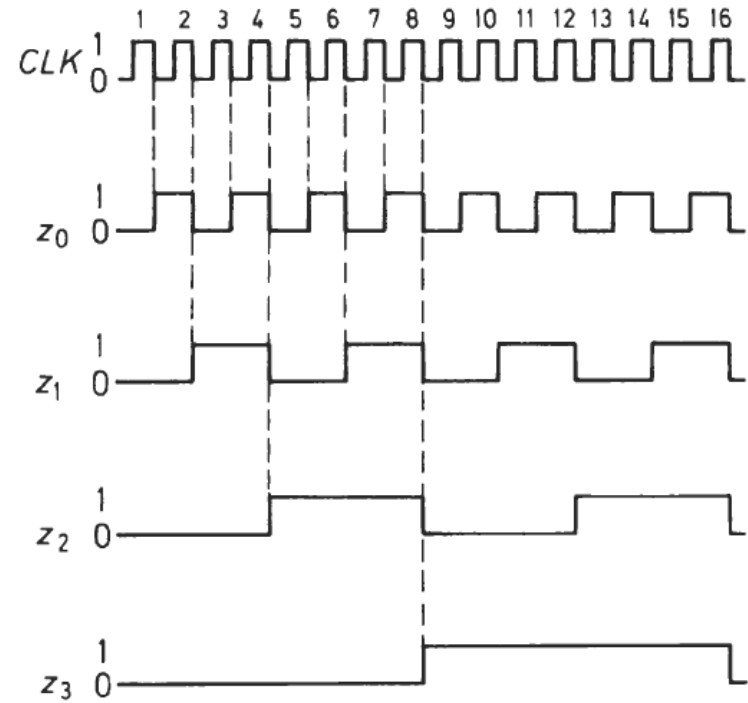
[1], p. 706



A Bus System



The Binary Counter I



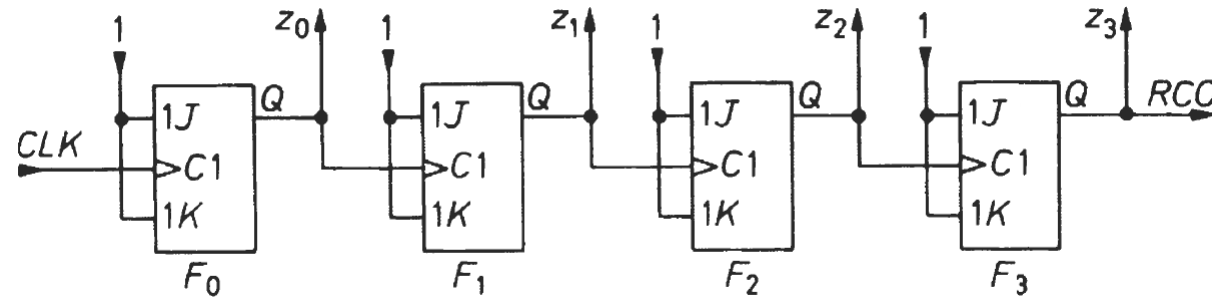
Z	z_3	z_2	z_1	z_0
	2^3	2^2	2^1	2^0
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1
16	0	0	0	0

[1], p. 694



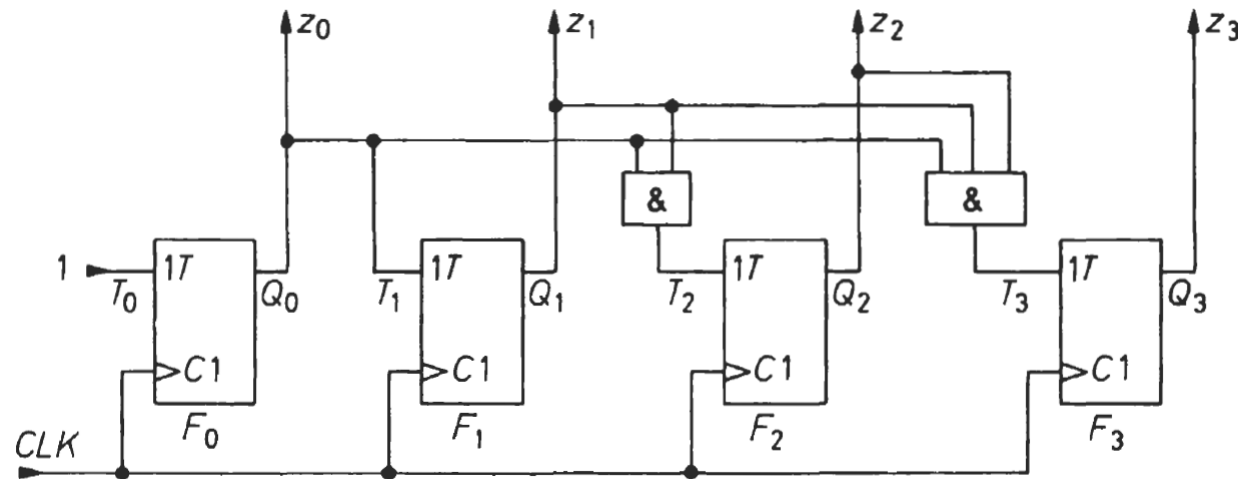
The Binary Counter II

- asynchronous:



[1], p. 694

- synchronous:



[1], p. 696



A Special Counter



References

- [1] Tietze, Ulrich; Schenk, Christoph (1999): Halbleiterschaltungstechnik. 11. Auflage. Berlin: Springer
- [2] Woitowitz, Roland; Urbanski, Klaus; Gehrke, Winfried (2012): Digitaltechnik. Ein Lehr- und Übungsbuch. 6. bearbeitete Auflage. Berlin: Springer



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